# Teen Mental-Health Prediction Classification Project

In [99]:

```
from IPython.display import Image,display
display(Image(filename="teen.jpg"))
```

In [100]:

```
#Import required libraries:
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

import time

from xgboost import XGBClassifier
from sklearn.model_selection import train_test_split
from sklearn.ensemble import RandomForestClassifier
from sklearn.metrics import accuracy_score,classification_report,ConfusionMatrixDisplay
from sklearn.tree import DecisionTreeClassifier, plot_tree
from sklearn.impute import KNNImputer
from sklearn.preprocessing import LabelEncoder
from scipy.stats import zscore
from imblearn.over_sampling import SMOTE,BorderlineSMOTE,ADASYN
from sklearn.model_selection import GridSearchCV



#Ignore all incoming warnings (if any):
import warnings
warnings.filterwarnings("ignore")


#Import the dataset:
df = pd.read_csv("dataset.csv")
```

In [101]:

```
#Preview the data:
df.sample(5)
```

Out[101]:

| | age | gender | daily_social_media_hours | platform_usage | sleep_hours | screen_time_before_sleep | academic_performance |
|---|---|---|---|---|---|---|---|
| **1187** | 18 | male | 2.0 | Instagram | 8.9 | 0.8 | 2.96 |
| **847** | 15 | female | 4.2 | TikTok | 6.8 | 1.4 | 2.01 |
| **681** | 19 | male | 6.8 | Instagram | 8.4 | 2.1 | 3.15 |
| **785** | 19 | female | 5.0 | Instagram | 6.3 | 2.9 | 2.04 |
| **786** | 19 | female | 4.2 | Instagram | 7.1 | 2.5 | 3.91 |

In [102]:

```
#Shape of the data:
df.shape
```

Out[102]:

```
(1200, 13)
```

In [103]:

```
#Detailed info about the dataset:
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1200 entries, 0 to 1199
Data columns (total 13 columns):
 #   Column                     Non-Null Count  Dtype
---  ------                     --------------  -----
 0   age                        1200 non-null   int64
 1   gender                     1200 non-null   object
 2   daily_social_media_hours   1200 non-null   float64
 3   platform_usage             1200 non-null   object
 4   sleep_hours                1200 non-null   float64
 5   screen_time_before_sleep   1200 non-null   float64
 6   academic_performance       1200 non-null   float64
 7   physical_activity          1200 non-null   float64
 8   social_interaction_level   1200 non-null   object
 9   stress_level               1200 non-null   int64
 10  anxiety_level              1200 non-null   int64
 11  addiction_level            1200 non-null   int64
 12  depression_label           1200 non-null   int64
dtypes: float64(5), int64(5), object(3)
memory usage: 122.0+ KB
```

In [104]:

```
#No. of null values in each attribute:
df.isnull().sum()
```

Out[104]:

```
age                          0
gender                       0
daily_social_media_hours     0
platform_usage               0
sleep_hours                  0
screen_time_before_sleep     0
academic_performance         0
physical_activity            0
social_interaction_level     0
stress_level                 0
anxiety_level                0
addiction_level              0
depression_label             0
dtype: int64
```

In [105]:

```
#View a statistical summary of the dataset:
df.describe()
```

Out[105]:

| | age | daily_social_media_hours | sleep_hours | screen_time_before_sleep | academic_performance | physical_activity |
|---|---|---|---|---|---|---|
| **count** | 1200.000000 | 1200.000000 | 1200.000000 | 1200.000000 | 1200.000000 | 1200.000000 |
| **mean** | 15.928333 | 4.536667 | 6.449417 | 1.740333 | 2.990383 | 1.014500 |
| **std** | 2.021947 | 2.029599 | 1.442677 | 0.716660 | 0.576758 | 0.582185 |

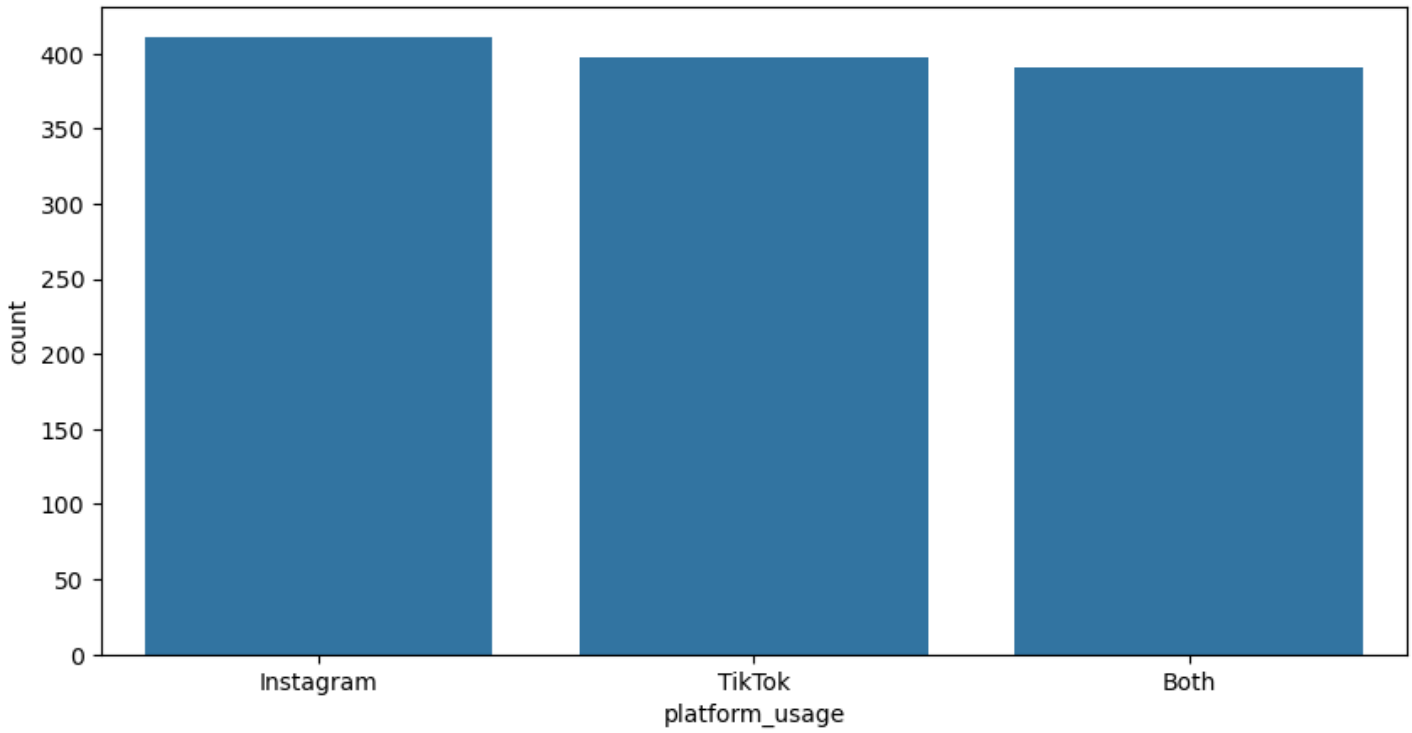
| | age | daily_social_media_hours | sleep_hours | screen_time_before_sleep | academic_performance | physical_activity |
|---|---|---|---|---|---|---|
| min | 13.000000 | 1.000000 | 4.000000 | 0.500000 | 2.000000 | 0.000000 |
| 25% | 14.000000 | 2.800000 | 5.200000 | 1.100000 | 2.500000 | 0.500000 |
| 50% | 16.000000 | 4.500000 | 6.500000 | 1.800000 | 2.990000 | 1.000000 |
| 75% | 18.000000 | 6.300000 | 7.600000 | 2.400000 | 3.480000 | 1.500000 |
| max | 19.000000 | 8.000000 | 9.000000 | 3.000000 | 4.000000 | 2.000000 |

In [106]:

```
#Social Media Platforms:
plt.figure(figsize=(10,5))
sns.countplot(x="platform_usage",data=df)
```

Out[106]:

```
<Axes: xlabel='platform_usage', ylabel='count'>
```




In [107]:

```
X = df.drop(["depression_label", "stress_level"], axis=1)
y = df["depression_label"]

#Split the data into training and testing set:
X_train , X_test , y_train , y_test = train_test_split(X,y,test_size=0.2,random_state=42
,stratify=y)
```

In [108]:

```
# Evidence for using SMOTE
print("Before SMOTE:")
print(y_train.value_counts())


# Fix the imbalance in the training data using SMOTENC
from imblearn.over_sampling import SMOTENC

# Categorical columns in X
cat_cols = ["gender", "platform_usage", "social_interaction_level"]

# Get their column positions
cat_indices = [X_train.columns.get_loc(col) for col in cat_cols]

# Apply SMOTENC
smote_nc = SMOTENC(categorical_features=cat_indices, random_state=42, k_neighbors=2)
```

```
X_trainsmote, y_trainsmote = smote_nc.fit_resample(X_train, y_train)


X_trainsmote_encoded = pd.get_dummies(X_trainsmote, drop_first=True)
X_test_encoded = pd.get_dummies(X_test, drop_first=True)

# Make both datasets have same columns
X_trainsmote_encoded, X_test_encoded = X_trainsmote_encoded.align(
    X_test_encoded, join="left", axis=1, fill_value=0
)

print("After SMOTE:")
print(y_trainsmote.value_counts())
```

```
Before SMOTE:
depression_label
0    935
1     25
Name: count, dtype: int64
After SMOTE:
depression_label
0    935
1    935
Name: count, dtype: int64
```

In [109]:

```
# Decision Tree Classifier
dec_model = DecisionTreeClassifier(max_depth=10, random_state=42)
dec_model.fit(X_trainsmote_encoded, y_trainsmote)

# Predictions
train_predictions = dec_model.predict(X_trainsmote_encoded)
test_predictions = dec_model.predict(X_test_encoded)

# Accuracy
accuracy_on_train = accuracy_score(y_trainsmote, train_predictions)
accuracy_on_test = accuracy_score(y_test, test_predictions)

print("Training Accuracy:", accuracy_on_train)
print("Testing Accuracy:", accuracy_on_test)
print("\nClassification Report:\n")
print(classification_report(y_test, test_predictions))
```

```
Training Accuracy: 0.990909090909091
Testing Accuracy: 0.95

Classification Report:

              precision    recall  f1-score   support

           0       0.99      0.96      0.97       234
           1       0.25      0.50      0.33         6

    accuracy                           0.95       240
   macro avg       0.62      0.73      0.65       240
weighted avg       0.97      0.95      0.96       240
```

In [110]:

```
results = {}
trained_models = {}

for name, model in models.items():
    model.fit(X_trainsmote_encoded, y_trainsmote)
    pred = model.predict(X_test_encoded)

    acc = accuracy_score(y_test, pred)

    results[name] = acc
    trained_models[name] = model
```

```
# Get best model
best_model_name = max(results, key=results.get)
best_model = trained_models[best_model_name]

print("Best Model:", best_model_name)
print("Best Accuracy:", results[best_model_name])
```

```
Best Model: Random Forest
Best Accuracy: 0.9666666666666667
```

In [111]:

```
models = {
    "Logistic Regression": LogisticRegression(),
    "KNN": KNeighborsClassifier(n_neighbors=5),
    "Naive Bayes": GaussianNB(),
    "Decision Tree": DecisionTreeClassifier(max_depth=5),
    "Random Forest": RandomForestClassifier(),
}

for name, model in models.items():
    model.fit(X_trainsmote_encoded, y_trainsmote)
    pred = model.predict(X_test_encoded)
    cr = classification_report(y_test, pred)
    print(f"{name}:\n {cr}")
    print("\n\n\n")
```

```
Logistic Regression:
               precision    recall  f1-score   support

           0       1.00      0.95      0.97       234
           1       0.29      0.83      0.43         6

    accuracy                           0.95       240
   macro avg       0.64      0.89      0.70       240
weighted avg       0.98      0.95      0.96       240




KNN:
               precision    recall  f1-score   support

           0       1.00      0.94      0.97       234
           1       0.26      0.83      0.40         6

    accuracy                           0.94       240
   macro avg       0.63      0.89      0.68       240
weighted avg       0.98      0.94      0.95       240




Naive Bayes:
               precision    recall  f1-score   support

           0       1.00      0.95      0.97       234
           1       0.29      0.83      0.43         6

    accuracy                           0.95       240
   macro avg       0.64      0.89      0.70       240
weighted avg       0.98      0.95      0.96       240




Decision Tree:
               precision    recall  f1-score   support
```

```
           0       0.99      0.95      0.97       234
           1       0.27      0.67      0.38         6

    accuracy                           0.95       240
   macro avg       0.63      0.81      0.68       240
weighted avg       0.97      0.95      0.96       240




Random Forest:
               precision    recall  f1-score   support

           0       0.98      0.98      0.98       234
           1       0.33      0.33      0.33         6

    accuracy                           0.97       240
   macro avg       0.66      0.66      0.66       240
weighted avg       0.97      0.97      0.97       240
```

In [112]:

```python
# Cross Validation
from sklearn.model_selection import KFold, cross_validate

kf = KFold(n_splits=5, shuffle=True, random_state=42)

scoring = {
    "accuracy":"accuracy",
    "precision": "precision",
    "recall": "recall",
    "f1": "f1"
}

results = []

for name, model in models.items():
    cv_results = cross_validate(
        model,
        X_trainsmote_encoded,
        y_trainsmote,
        cv=kf,
        scoring=scoring
    )

    accuracy = np.mean(cv_results["test_accuracy"])
    precision = np.mean(cv_results["test_precision"])
    recall = np.mean(cv_results["test_recall"])
    f1_score = np.mean(cv_results["test_f1"])

    results.append({
        "Model Name": name,
        "Accuracy": round(accuracy, 2),
        "Precision": round(precision, 2),
        "Recall": round(recall, 2),
        "F1-Score": round(f1_score, 2),
    })

results_df = pd.DataFrame(results).sort_values(by="F1-Score", ascending=False)
print(results_df)
```

```
            Model Name  Accuracy  Precision  Recall  F1-Score
4        Random Forest      0.99       0.98    1.00      0.99
3        Decision Tree      0.98       0.96    0.99      0.98
2          Naive Bayes      0.97       0.95    0.99      0.97
```

```
1                   KNN       0.96        0.92     1.00       0.96
0   Logistic Regression       0.94        0.92     0.98       0.95
```

In [113]:

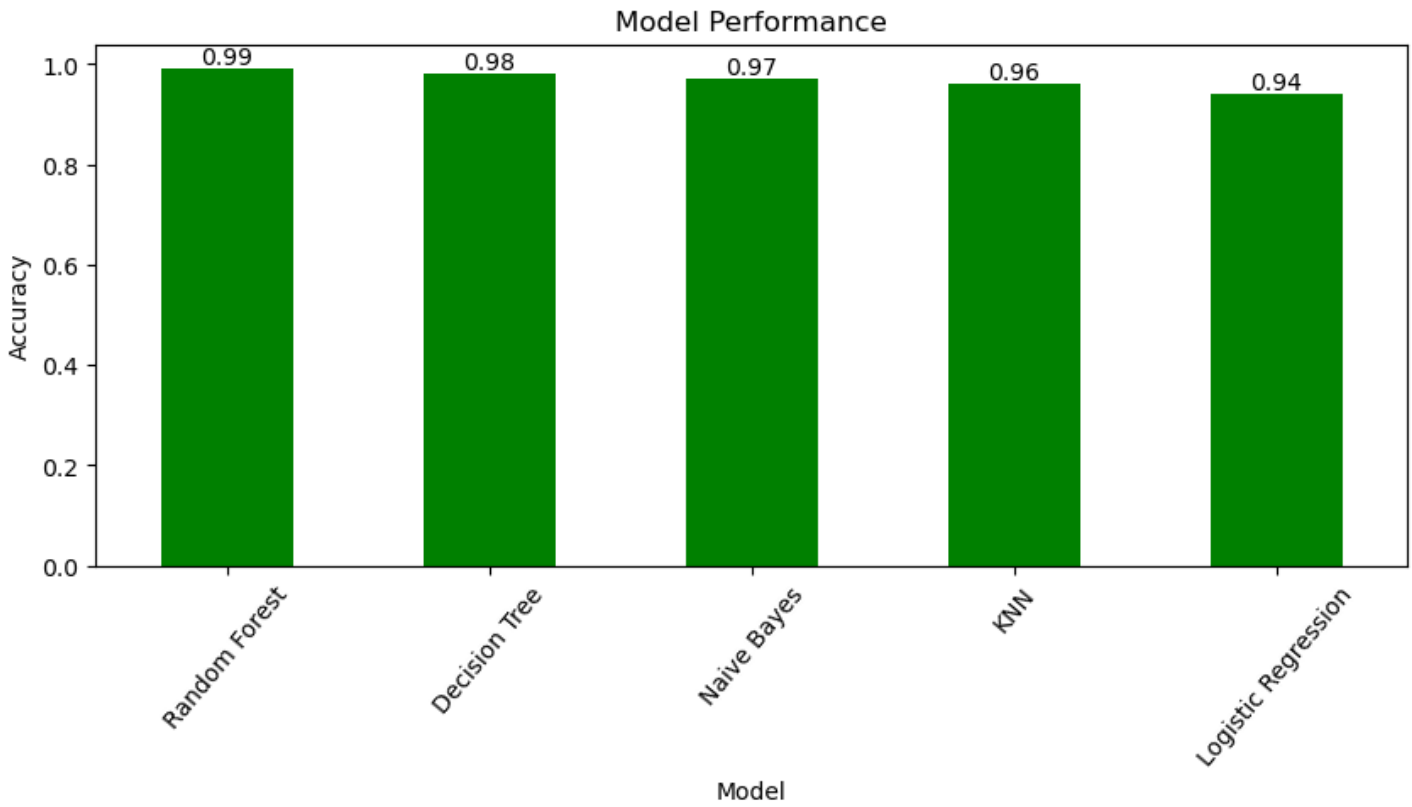
```
plt.figure(figsize=(10, 4))

ax = results_df.set_index("Model Name")["Accuracy"].plot(kind='bar',color="green")

plt.title("Model Performance")
plt.xlabel("Model")
plt.ylabel("Accuracy")
plt.xticks(rotation=50)

# Add labels on each bar
for i, v in enumerate(results_df["Accuracy"]):
    ax.text(i, v + 0.01, str(v), ha='center')

plt.show()
```




In [ ]:

In [114]:

```
best_model
```

Out[114]:

```
▾
 RandomForestClassifier
                      i ?
RandomForestClassifier()
```

In [115]:

```
#Export the best performing model:
import joblib
joblib.dump(best_model,"teen_mental_health_prediction_model.pkl")
```

Out[115]:

```
['teen_mental_health_prediction_model.pkl']
```

```
['teen_mental_health_prediction_model.pkl']
```

In [ ]:

```

```

## Project made by: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In [ ]:

```

```